PAPER_935: GW170817 Tidal Deformability Phonon Correction

Author: Daniel T. Murphy – Star Magic / UQFF Framework **Date:** 2026-04-12 **Session:** 212 **Source:** ns_phonon_gw170817_wstp.py (TidalDeformabilityPhononCorrection) **Calculator:** GW170817TidalDeformabilityPhononCalc (CP4 #519) **CVW:** v2.0.0 compliant

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Abstract

We compute the phonon-corrected tidal deformability Lambda_UQFF for GW170817 neutron stars. The UQFF phonon coupling modifies the Love number k_2 through the SCm lattice interaction, yielding Lambda_UQFF = Lambda_GR (1 + Phi S_26 D_total). The UQFF-predicted range Lambda in [190, 600] is consistent with the LIGO constraint Lambda_tilde < 800 and provides tighter bounds than GR alone.

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1. Core Equations

UQFF-corrected tidal deformability:

$$\Lambda_{\text{UQFF}} = \Lambda_{\text{GR}} \cdot (1 + \Phi \cdot S_{26} \cdot D_{\text{total}})$$

where: - Λ_{GR} is the GR tidal deformability from the neutron star equation of state - Φ is the phonon flux at SCm resonance frequency - $S_{26} = \sum_{k=1}^{26} e^{-[\text{SSq}] \cdot k/26}$ is the 26-layer suppression sum - $D_{\text{total}} = 0.333$ is the phonon suppression factor

LIGO constraint:

$$\tilde{\Lambda} = \frac{16}{13} \frac{(m_1 + 12m_2)m_1^4\Lambda_1 + (m_2 + 12m_1)m_2^4\Lambda_2}{(m_1 + m_2)^5} < 800$$

UQFF Range

Regime	Lambda_UQFF
Soft EOS (Lambda_GR ~ 150)	~190
Moderate EOS (Lambda_GR ~ 400)	~500
Stiff EOS (Lambda_GR ~ 500)	~600

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2. UQFF Integration

The `GW170817TidalDeformabilityPhononCalc` (CP4 #519) takes Lambda_GR, D_total, and [SSq] as inputs. It checks both the LIGO bound (Lambda < 800) and the UQFF-predicted range [190, 600]. The simulate() method sweeps Lambda_GR = [100, 200, 300, 400, 500].

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3. Physical Significance

Tidal deformability is a direct probe of the neutron star equation of state. The UQFF phonon correction arises because crust phonon modes modify the tidal response: the SCm lattice coupling adds a rigidity contribution to k_2 that stiffens the effective EOS. This narrows the allowed Lambda range from the broad GR prediction to the UQFF band [190, 600], which future detections (GW230529, O5 run) can test.

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4. Source Data

- **File:** ns_phonon_gw170817_wstp.py
- **Session:** 212
- **CP4 Class:** GW170817TidalDeformabilityPhononCalc (#519)

References

1. Murphy, D.T. – Star Magic UQFF Framework (2024-2026)
2. Abbott, B.P. et al. – GW170817: Measurements of Neutron Star Radii and EOS, PRL 121, 161101 (2018)
3. Hinderer, T. – Tidal Love numbers of neutron stars, ApJ 677, 1216 (2008)